

Homework 2

(due April 6 @15:55, in Blackboard)

1) For all possible cases of $P \in \mathbf{S}^d, q \in \mathbf{R}^d$:

$$\text{minimize } x^T P x + q^T x$$

2) Problem 4.8 (a)-(d) from the reference book.

3) Problem 4.21 from the reference book.

4) Solve the following optimization problems in CVXPY. Your solution should include both the optimal value and the optimal decision variables, as well as the scripts you used to get them.

a)

$$\begin{aligned} &\text{minimize} && x_1 + 2x_2 + 3x_3 - x_4 \\ &\text{subject to} && x_1 - x_2 + x_3 - 2x_4 \leq 6 \\ &&& 2x_1 + x_2 - x_3 \geq 2 \\ &&& -x_1 + x_2 - x_3 + x_4 \geq 8 \\ &&& 0 \leq x_1 \leq 3 \\ &&& 1 \leq x_2 \leq 4 \\ &&& 0 \leq x_3 \leq 10 \\ &&& 2 \leq x_4 \leq 5 \end{aligned}$$

b)

$$\begin{aligned} &\text{minimize} && 9x_1^2 + 9x_2^2 - 30x_1 - 72x_2 \\ &\text{subject to} && -2x_1 - x_2 \geq -4 \\ &&& x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

c)

$$\begin{aligned} &\text{minimize} && t \\ &\text{subject to} && \begin{bmatrix} 1 & 0 & 7 \\ 0 & 5 & 3 \\ 7 & 3 & 2 \end{bmatrix} + x_1 \begin{bmatrix} 2 & -2 & 2 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} + x_2 \begin{bmatrix} 9 & 2 & 1 \\ 2 & 5 & 6 \\ 1 & 6 & 4 \end{bmatrix} \preceq tI \end{aligned}$$

- 5) *Linear regression.* The California Housing dataset contains 20,640 instances with 8 attributes and 1 target. We provide two files for training and two for testing: a) ‘train_data.csv’ contains the attributes and ‘train_target.csv’ contains the targets for 16,640 instances (respectively denoted as $A \in \mathbf{R}^{16,640 \times 8}$, $b \in \mathbf{R}^{16,640}$); b) ‘test_data.csv’ and ‘test_target.csv’ contain the attributes and targets for the remaining 4,000 instances ($A_{\text{test}} \in \mathbf{R}^{4,000 \times 8}$, $b_{\text{test}} \in \mathbf{R}^{4,000}$). The goal is to obtain a model $x \in \mathbf{R}^8$ with low test error $\|A_{\text{test}}x - b_{\text{test}}\|_2^2$, using only the training data A, b .

Consider the following three alternatives:

- Least squares (LS):

$$\text{minimize} \quad \|Ax - b\|_2^2$$

- LASSO ($\lambda > 0$):

$$\text{minimize} \quad \|Ax - b\|_2^2 + \lambda \|x\|_1$$

- Ridge regression ($\lambda > 0$):

$$\text{minimize} \quad \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

- Obtain the least squares solution x_{LS} .
- Solve LASSO for various choices of λ and plot $(\|Ax - b\|_2^2, \|x\|_1)$ for each choice. What do you observe? Pick your ‘best’ solution as x_{LASSO} .
- Repeat the same as in b) to obtain x_{Ridge} .
- Evaluate $\|A_{\text{test}}x - b_{\text{test}}\|_2^2$ for the three solutions you obtained in a), b), c), and discuss your findings (please share your plots and scripts).

The following problem is for *extra credit*.

- Consider the problem of minimizing a convex differentiable function on $\{x \in \mathbf{R}^n \mid x \succeq 0, \mathbf{1}^T x = 1\}$.
 - Show that the necessary and sufficient condition for optimality is

$$\left\{ x_i > 0 \implies \frac{\partial f}{\partial x_i} = \min_{j=1, \dots, n} \frac{\partial f}{\partial x_j} \right\} \cap \{x \succeq 0, \mathbf{1}^T x = 1\}.$$
 - For the case $f(x) := \frac{1}{2} \|x - y\|_2^2$ (where $y \in \mathbf{R}^n$ is given) derive an efficient algorithm for computing the optimal solution and analyze its complexity.
 - For a randomly chosen $y \in \mathbf{R}^{500}$, solve the problem using your algorithm as well as via CVXPY. Compare and explain the implications (please share your results and scripts).